

#This code will be used to demonstrate theorem 6.18. We will demonstrate that $Q1 = \frac{x^2}{s^2}$ is increasing along the γ_2 curve; (Part 3 — calculating the sign of the derivative of $Q1'(\gamma_2(1), 1)$)

#Define Q1;

$$Q1 := \frac{x^2 \cdot ((z + w)^2 - y^2)}{y^2 \cdot z \cdot w} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \left. \right\} :$$

$q1 := \text{simplify}(\text{expand}(\text{subs}(\text{parametrization}, Q1))) ;$

$$q1 := \frac{4(b+a)^2(ab-1)^2}{(a^2+1)(b^2+1)(a-b)^2} \quad (1)$$

#Define r2

$$\begin{aligned} r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\ & (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\ & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\ & + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\ & - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\ & - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\ & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\ & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\ & \left. a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \right) \end{aligned}$$

$$\begin{aligned}
& -27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\
& - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\
& - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\
& + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
& - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\
& - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
& + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
& - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
& - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& \left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#Calculating the derivative of $Q1$ at $b=1$

$$d := \text{eval}(\text{diff}(q1, a), b = 1) \cdot \left(-\frac{\text{eval}(\text{diff}(r2, b), b = 1)}{\text{eval}(\text{diff}(r2, a), b = 1)} \right) + \text{eval}(\text{diff}(q1, b), b = 1) :$$

#Locate a range for $\gamma_2(1)$

$$\text{sturm}(\text{sturmseq}(\text{eval}(r2, b = 1), a), a, 2.6, 3); \text{\#number of roots in } [2.6, 3)$$

1

(2)

#Define nd

$$nd := \text{numer}(d)$$

$$\begin{aligned} nd := & 2(1+a)(a^{35} + 7a^{34} + 7a^{32} - 6a^{31} - 50a^{30} - 206a^{29} - 1842a^{28} + 1070a^{27} \\ & + 24634a^{26} + 184242a^{25} - 12614a^{24} - 40046a^{23} - 2487082a^{22} - 489830a^{21} \\ & - 7020714a^{20} - 480768a^{19} - 9170088a^{18} + 139962a^{17} - 7904168a^{16} + 607054a^{15} \\ & - 4536054a^{14} + 216918a^{13} - 1597686a^{12} + 14354a^{11} - 343962a^{10} - 117482a^9 \\ & + 22758a^8 - 36058a^7 - 2478a^6 + 862a^5 - 814a^4 - 65a^3 + a^2 - 2a + 1) \end{aligned}$$

(3)

#Define dd

$$dd := \text{denom}(d)$$

$$\begin{aligned} dd := & (a^2 + 1)^2 a^3 (a^{28} - 7a^{24} - 208a^{22} + 405a^{20} + 23552a^{19} + 1232a^{18} + 10752a^{17} \\ & - 319075a^{16} - 48640a^{15} - 592416a^{14} - 26112a^{13} - 570845a^{12} + 23040a^{11} - 398944a^{10} \\ & + 19968a^9 - 143253a^8 + 5632a^7 - 42000a^6 - 4608a^5 + 647a^4 - 3584a^3 + 144a^2 \\ & - 65)(a - 1) \end{aligned}$$

(4)

#Verify that nd and dd have the same sign in the interval.

$$\text{sturm}(\text{sturmseq}(nd, a), a, 2.6, 3); \text{\#number of roots in } [2.6, 3) ;$$

$$\text{sturm}(\text{sturmseq}(dd, a), a, 2.6, 3); \text{\#number of roots in } [2.6, 3)$$

0

0

(5)

#Calculate the sign of $Q1'(\gamma_2(1), 1)$ using $Q1'(3, 1)$

$$\frac{\text{eval}(nd, a = 3)}{\text{eval}(dd, a = 3)}$$

$$\frac{6619774424288}{564267245625}$$

(6)